Bank FSM v0.1

State Definitions, Request Cases, Edge Cases & FSM Diagrams

IDLE · OPEN · READ · WRITE · PRECHARGE · REF · RFM · Power States · Timer Ownership

Version	v0.1	**States (v0.1)**	IDLE, OPEN	**States (full)**	11
Instances	16 (one per bank)	**Key principle**	Intent / Legality / Selection separation	**Standard**	JESD79-5

1 Design Principle

Intent / Legality / Selection — Core Separation

Bank FSM = *Intent* — what the bank needs to do next
Timing Control = *Legality* — when is it legal (`cmd_ok`)
Scheduler = *Selection* — which bank wins this cycle

The Bank FSM never directly issues a command. It asserts `need_ACT`, `need_RD`, `need_WR`, `need_PRE`. The Scheduler dispatches only when Timing Control's `cmd_ok` is also asserted. These three signals must all be true simultaneously for a command to be issued.

2 Bank Context (v0.1)

Field	Width	Description
`bank_state`	1b (v0.1)	Current FSM state: IDLE or OPEN
`open_row`	17b	Row address currently activated in this bank
`tRAS_timer`	9b	Countdown from ACT issue. Guards PRE.
`tRCD_timer`	8b	Countdown from ACT issue. Guards first RD/WR.
`tRP_timer`	8b	Countdown from PRE issue. Guards next ACT.
`tWR_timer`	8b	Countdown from WR burst end. Guards PRE.
`tRTP_timer`	6b	Countdown from RD issue. Guards PRE.

Extended Context (full version)

Additional fields needed beyond v0.1: `tRC_timer` (ACT-to-ACT same bank), `tRFC_timer` (after REFab), `tRFCsb_timer` (after REFsb), `tXS_timer` (after SRX), `tXP_timer` (after PDX), `rfm_raa_counter` (per-bank RAA for RFM).

3 Top-Level Bank FSM (v0.1)

4 READ Request Cases

4.1 Case R1 — Row Hit

Condition: `bank_state == OPEN AND open_row == req.row`

Timers involved: tCCD_L (same BG) / tCCD_S (diff BG) / tWTR_L/S (if previous WR) / tRTW (if switching RD→WR mode irrelevant here but checked globally)

4.2 Case R2 — Row Closed (Bank IDLE)

Condition: `bank_state == IDLE`

Timers involved: tRRD_L/S (cross-bank), tFAW (4-ACT window), tRCD (bank private)

4.3 Case R3 — Row Miss

Condition: `bank_state == OPEN AND open_row != req.row`

Timers involved: tRAS (bank private, must expire before PRE), tRP (bank private), tRRD_L/S, tFAW, tRCD, tCCD_L/S

5 WRITE Request Cases

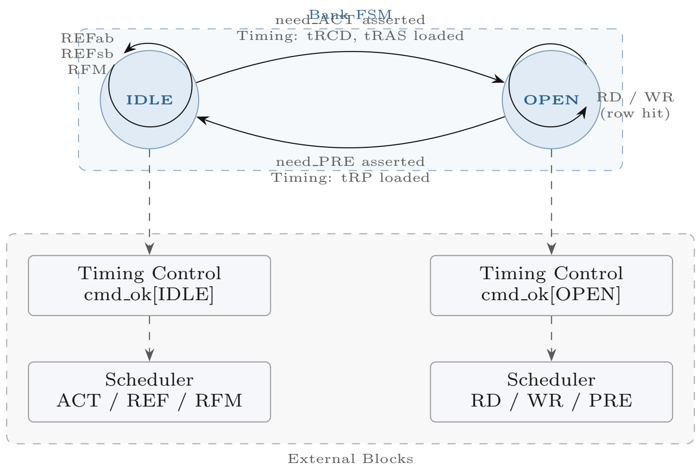

Figure 1. Bank FSM v0.1: two states. All intent outputs (need_ACT etc.) gated by Timing Control before the Scheduler dispatches.

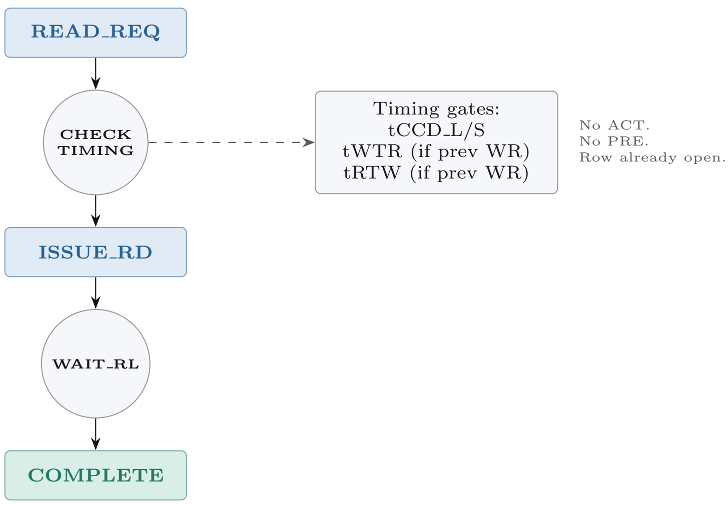

Figure 2. R1: Row hit. CAS issued directly. No ACT/PRE. Timing Control must clear tCCD, tWTR, tRTW before dispatch.

5.1 Case W1 — Write Hit

Condition: `bank_state == OPEN` AND `open_row == req.row`

Timers involved: tCCD_L_WR (same BG WR→WR), tCCD_L_WR2 (same BG, 2nd not RMW), tCCD_S_WR (diff BG), tRTW (if previous RD)

5.2 Case W2 — Row Closed

Condition: `bank_state == IDLE`

5.3 Case W3 — Row Miss

Condition: `bank_state == OPEN` AND `open_row != req.row`

6 Maintenance Cases (REF, RFM, Power)

6.1 Case M1 — All-Bank Refresh (REFab)

Condition: Refresh Scheduler asserts `ref_req_ab`. All 16 banks receive simultaneously.

6.2 Case M2 — Same-Bank Refresh (REFsb)

Condition: Refresh Scheduler asserts `ref_req_sb[ba]`. Only target BA locked.

6.3 Case M3 — RFM (Rowhammer Management)

Condition: RFM FSM asserts `rfm_req[b]` when `RAA[b]` $\leq$ `RAAIMT`.

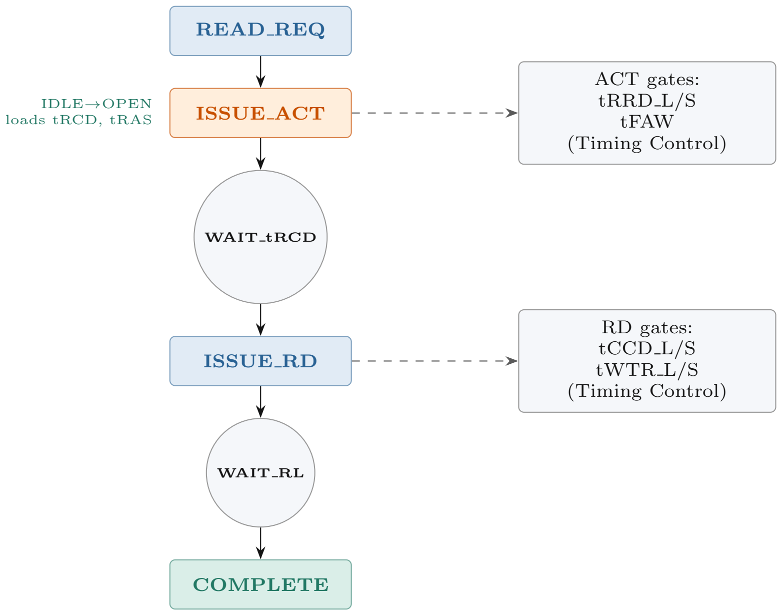

Figure 3. R2: Bank idle. ACT first, wait tRCD, then RD. ACT also blocked by Timing Control until tRRD and tFAW satisfied.

6.4 Case M4 — ZQ Calibration

Condition: ZQcal FSM fires periodic `tZQCS` timer or host trigger.

7 Power State Cases

7.1 Case P1 — Precharge Power Down

Condition: All banks in IDLE. CKE to be deasserted.

7.2 Case P2 — Active Power Down

Condition: Bank in OPEN. CKE deasserted while row active.

7.3 Case P3 — Self Refresh

Condition: All banks precharged. Deep power state.

7.4 Case P4 — MPSM (Maximum Power Saving Mode)

Condition: All banks precharged. MR2 OP[1:0]=01.

8 Timing Gate Cases (Scheduler Blocks)

These are not FSM states. They are legality checks in Timing Control that block the Scheduler from dispatching even when the Bank FSM has asserted intent.

Gate	Owner	Blocks	Constraint
tRRD_L	Timing Control	ACT (same BG)	max(8 nCK, 5 ns) between ACTs same BG
tRRD_S	Timing Control	ACT (diff BG)	8 nCK between ACTs different BG
tFAW	Timing Control	ACT (any bank)	Max 4 ACTs within tFAW window
tCCD_L	Timing Control	RD (same BG)	max(8 nCK, 5 ns) RD-to-RD same BG
tCCD_S	Timing Control	RD (diff BG)	8 nCK RD-to-RD diff BG
tCCD_L_WR	Timing Control	WR (same BG)	max(32 nCK, 20 ns) WR-to-WR same BG
tCCD_L_WR2	Timing Control	WR (same BG, not RMW)	max(16 nCK, 10 ns)
tWTR_L	Timing Control	RD after WR (same BG)	$WL + BL/2 + t_{WTR\_L}$
tWTR_S	Timing Control	RD after WR (diff BG)	$WL + BL/2 + t_{WTR\_S}$
tRTW	Timing Control	WR after RD (any)	$RL + BL/2 - WL + 2 + t_{WPRE}$

9 Timer Ownership Summary

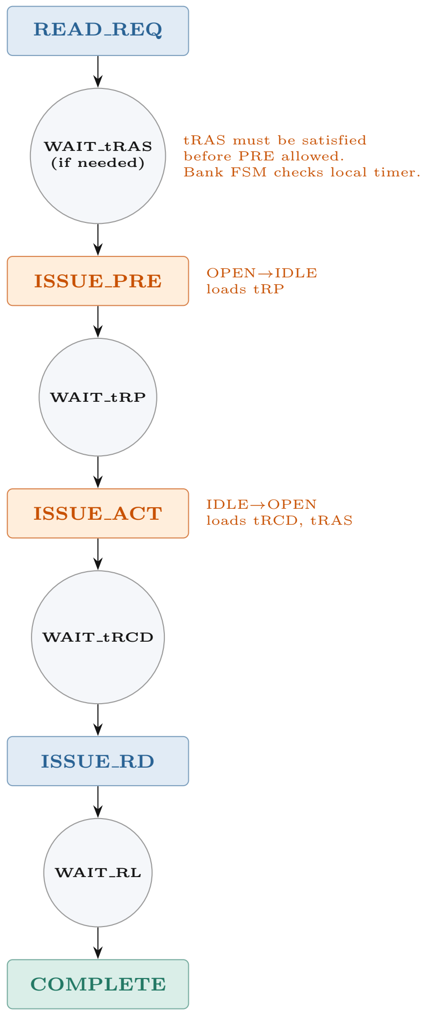

Figure 4. R3: Row miss. PRE current row, wait tRP, ACT new row, wait tRCD, RD. Longest read latency case.

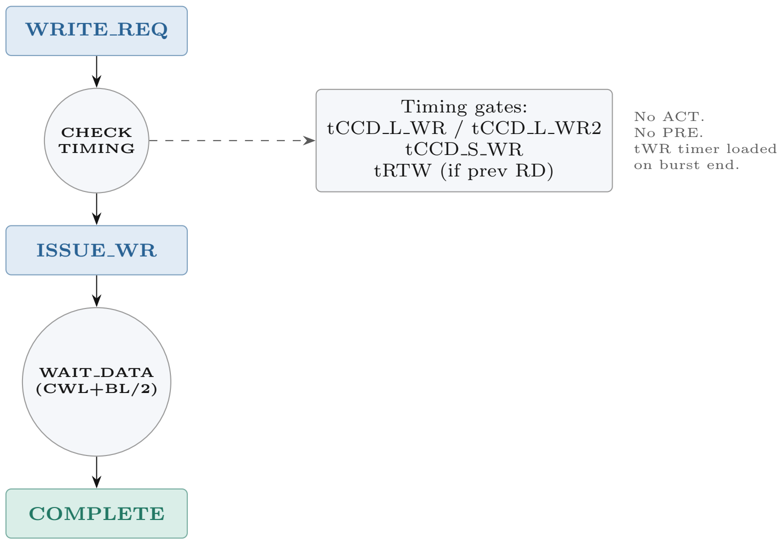

Figure 5. W1: Write hit. CAS(WR) directly. tCCD_L_WR applies (max 32 nCK same BG). tWR loaded at burst end for future PRE guard.

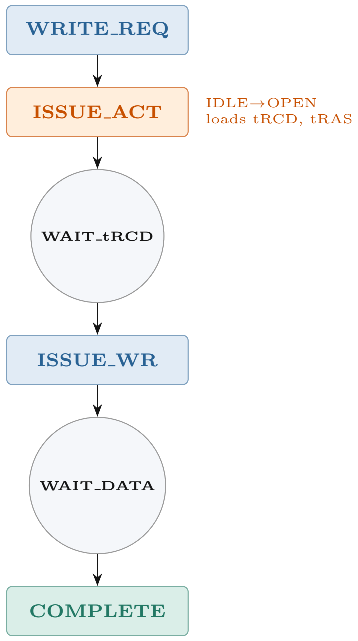

Figure 6. W2: Bank idle. ACT then WR. tRCD must elapse before WR can be issued.

Timer	Owner	Loaded By / Guards
tRAS	Bank FSM	ACT issued → PRE allowed only after expiry
tRCD	Bank FSM	ACT issued → first RD/WR on same bank
tRP	Bank FSM	PRE issued → next ACT on same bank
tWR	Bank FSM	WR burst end → PRE on same bank
tRTP	Bank FSM	RD issued → PRE on same bank
tRC	Bank FSM	ACT issued → next ACT same bank (= tRAS+tRP)
tRFC1	Bank FSM	REFab issued → all commands to rank
tRFCsb	Bank FSM	REFsb issued → commands to refreshed bank
tXP	Bank FSM	PDX → first command after power-down exit
tXS	Bank FSM	SRX → first ACT after self-refresh exit
tMRD	Bank FSM	MRW issued → next MRW
tRRD_L	Timing Control	Last ACT in same BG → next ACT same BG
tRRD_S	Timing Control	Last ACT globally → next ACT diff BG
tFAW	Timing Control	4-entry ACT ring buffer → 5th ACT blocked if oldest < tFAW
tCCD_L	Timing Control	Last CAS same BG → next RD/RD same BG
tCCD_S	Timing Control	Last CAS globally → next CAS diff BG
tCCD_L_WR	Timing Control	Last WR same BG → next WR same BG

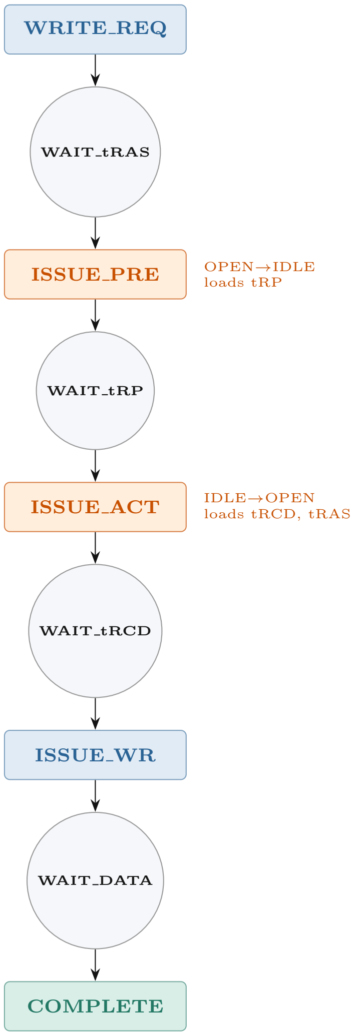

Figure 7. W3: Row miss. Same as R3 but CAS(WR) at end. tWR loaded at burst end.

10 Full Bank FSM (All States)

Expanded from v0.1 (IDLE/OPEN) to include all 11 states required for full DDR5 support.

11 Left-Over / Edge Cases

11.1 Auto-Precharge (RDA / WRA)

Condition: AP bit (CA10) set in RD or WR command.

- RDA: hidden PRE begins internally at tRTP after the CAS command. Bank transitions READING → PRECHARGING automatically, no explicit PRE needed.
- WRA: hidden PRE begins at CWL+BL/2 after WR. Bank transitions WRITING → PRECHARGING automatically.
- Bank FSM must not accept another ACT until tRP expires after auto-precharge.
- Useful for closed-page policy to avoid explicit PRE command overhead.

Command	Hidden PRE starts at	Guard for next ACT
RDA	tRTP after CAS	tRTP + tRP
WRA	CWL+BL/2 after WR	CWL+BL/2+tWR+tRP

11.2 tRAS Maximum Violation

Condition: Row has been open for $> 9 \times t_{REFI}$ with no PRE.

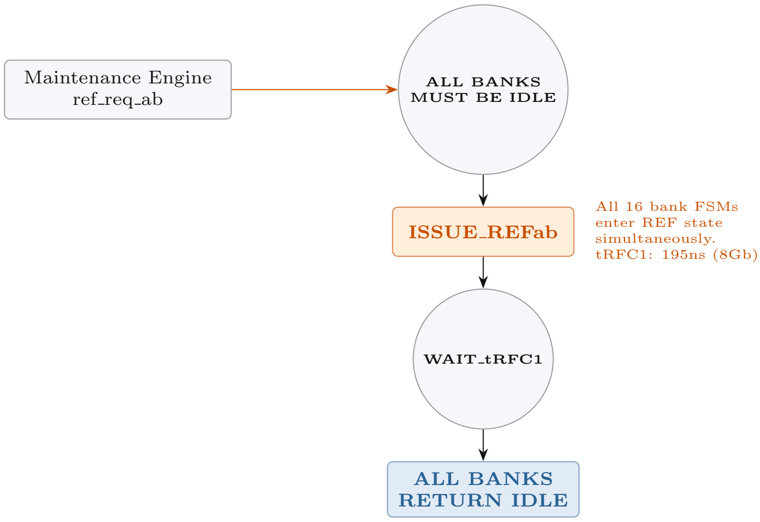

Figure 8. M1: REFab. All banks blocked for tRFC1. No other commands during window.

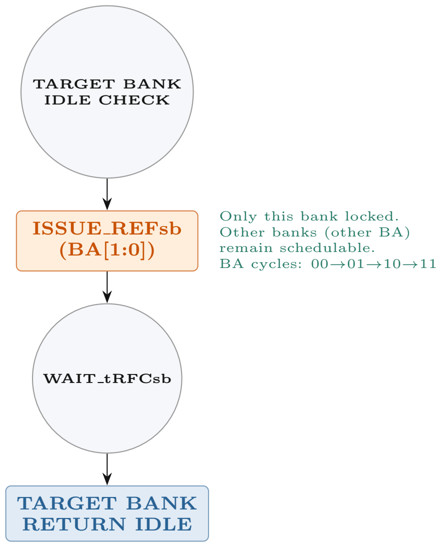

Figure 9. M2: REFsb. Only target bank locked for tRFCsb. DDR5 per-bank refresh advantage.

tRASmax not implemented in v0.1

JEDEC requires that a row may not remain open indefinitely. Maximum row active time $\approx 9 \times t_{REFI}$. Bank FSM must assert forced PRE if this timer expires. Not implemented in v0.1. Required for DDR5 compliance.
Required: `tRASmax_timer` in Bank Context. When expired: assert `need_PRE` regardless of scheduler state.

11.3 BST (Burst Terminate)

Condition: BC8 truncation of BL16 burst.

- BST command truncates the current burst at BC8 (8 UI instead of 16).
- Bank FSM must update burst-end timer to reflect shortened burst.
- tWR and tRTP timers loaded based on actual burst end, not full BL16.
- Affects: READING → PRECHARGING and WRITING → PRECHARGING transitions.

11.4 Rank Conflict Hazard

Condition: Command targets a bank in a rank that is currently refreshing.

- Bank FSM asserts `need_RD` or `need_WR`.
- Rank Manager (missing block) must assert `rank_busy[r]`.
- Timing Control's `gate_rfc[r]` blocks dispatch until tRFC1 expires.
- Resolution: Rank Manager block (currently missing, see CRITICAL-2).

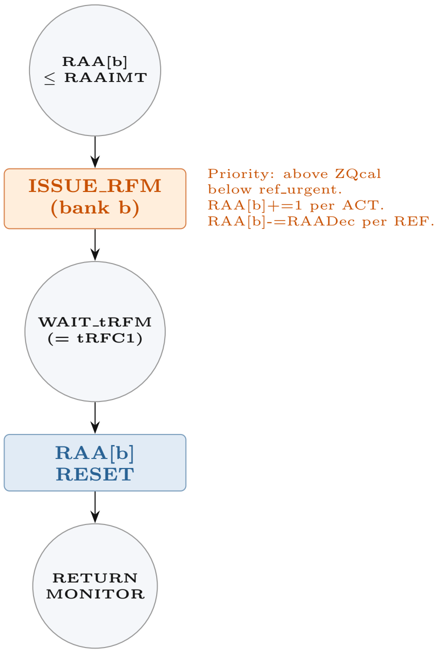

Figure 10. M3: RFM. Bank enters RFM_ACTIVE for tRFM. RAA counter reset after.

11.5 Multi-Rank Turnaround

Condition: Back-to-back commands to different ranks.

- Same-rank: normal timing constraints apply.
- Different ranks: ODT switching adds latency (`tADC`).
- CKE per-rank: power state transitions are independent per rank.
- Bank FSM is per-bank and unaware of other ranks — this is Rank Manager territory.

11.6 WR_PARTIAL (RMW)

Condition: Write with byte-mask; DRAM must perform Read-Modify-Write internally.

- Triggered by `wr_partial=1` in command packet.
- Bank FSM flow: same as W1/W2/W3 cases.
- Key difference: `tCCD_L_WR2` (not `tCCD_L_WR`) applies for the **next** WR to same BG, since the second write is an RMW.
- Timing Control must distinguish WR vs WR_PARTIAL to select correct tCCD variant.
- This disambiguation is currently unimplemented (see constraint checklist).

11.7 3DS (3D Stacked) Die Select

Condition: Target is a 3DS device with CID[3:0] die select.

- Bank FSM is per-bank per-die. CID field in command selects die.
- Additional timing: `tCCD_slr` (same logical rank, diff die) and `tCCD_dlr` (different logical rank).
- Bank FSM unchanged; CID routing is Rank Manager / Command Selector responsibility.

11.8 FGR (Fine Granularity Refresh) Mode Interaction

Condition: REF_MODE = FGR-2× or FGR-4×.

- tRFC changes: FGR-2× uses tRFC2; FGR-4× uses tRFC4.
- Bank FSM: tRFC timer loaded from correct register based on REF_MODE.
- REFsb bank address cycling: frequency doubles or quadruples.
- Budget counter threshold halved (FGR-2×) or quartered (FGR-4×).

11.9 Transition Legality Matrix

All valid and invalid state transitions for the full 11-state FSM:

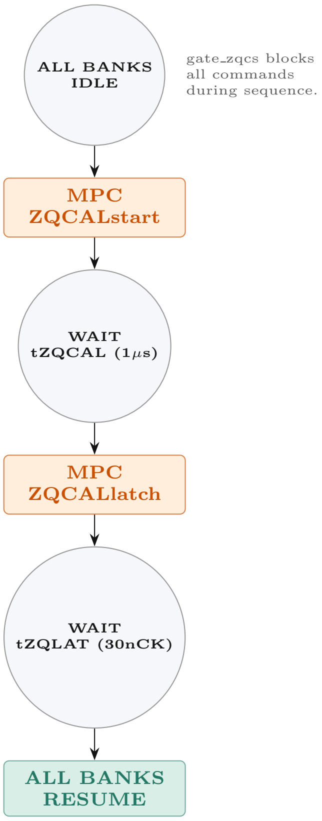

Figure 11. M4: ZQcal. Full rank stall for tZQCAL+tZQLAT. DRAM recalibrates driver/ODT impedance to RZQ.

From	To	Legal	Condition
IDLE	ACTIVATING	Yes	ACT received, tRRD/tFAW clear
IDLE	REF (ab)	Yes	REFab from Refresh Scheduler
IDLE	REF (sb)	Yes	REFsb targeting this BA
IDLE	RFM ACTIVE	Yes	RFM from RFM FSM
IDLE	POWER DOWN	Yes	PDE asserted
IDLE	SELF REFRESH	Yes	SRE (all banks must be IDLE)
ACTIVATING	BANK ACTIVE	Yes	tRCD expired
BANK ACTIVE	READING	Yes	RD/RDA command, tRCD clear
BANK ACTIVE	WRITING	Yes	WR/WRA command, tRCD clear
BANK ACTIVE	PRECHARGING	Yes	PRE/PREA, tRAS satisfied
BANK ACTIVE	POWER DOWN	Yes	PDE (active PD)
READING	BANK ACTIVE	Yes	Burst end, no AP
READING	PRECHARGING	Yes	RDA (auto-precharge) after tRTP
WRITING	BANK ACTIVE	Yes	Burst end, no AP
WRITING	PRECHARGING	Yes	WRA (auto-precharge) after tWR
PRECHARGING	IDLE	Yes	tRP expired
IDLE	BANK ACTIVE	**No**	Cannot skip ACTIVATING
BANK ACTIVE	ACTIVATING	**No**	Must PRE first
READING	WRITING	**No**	Direct, must complete burst
ACTIVATING	IDLE	**No**	Cannot abort activation
IDLE	READING	**No**	Must ACT first
IDLE	WRITING	**No**	Must ACT first

12 Bank FSM v0.1 vs Full Comparison

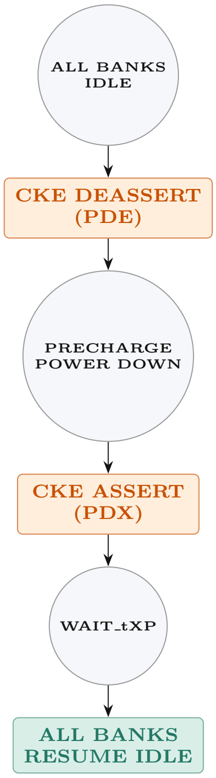

Figure 12. P1: Precharge power down. Entered from IDLE. tXP after PDX before commands.

Feature	v0.1	Full	Notes
States	2 (IDLE, OPEN)	11	ACTIVATING, READING, WRITING, PRECHARGING, REFab, REFsb, RFM, PD, SR
READ cases	3 (hit/closed/miss)	3 + RDA	Auto-precharge variant
WRITE cases	3 (hit/closed/miss)	3 + WRA	Auto-precharge + WR_PARTIAL (RMW)
Refresh	Not in v0.1	REFab + REFsb + FGR	Per-bank lock vs full-rank lock
RFM	Not in v0.1	Full RAA tracking	Per-bank counter, RAAIMT threshold
Power states	Not in v0.1	PD (pre/active) + SR + MPSM	CKE-controlled
tRASmax guard	Not implemented	Required	Force-PRE after $9 \times t_{REFI}$
BST handling	Not in v0.1	Required	BC8 burst terminate affects timers
3DS CID	Not in v0.1	Required	tCCD_slr / tCCD_dlr
WR_PARTIAL	Not in v0.1	Required	RMW tCCD disambiguation

Bank FSM v0.1 specification. JEDEC JESD79-5 (DDR5). 16 instances, one per bank. Key principle: Bank FSM = Intent, Timing Control = Legality, Scheduler = Selection. tRASmax, BST, WR_PARTIAL, 3DS, and FGR edge cases deferred to v0.2.

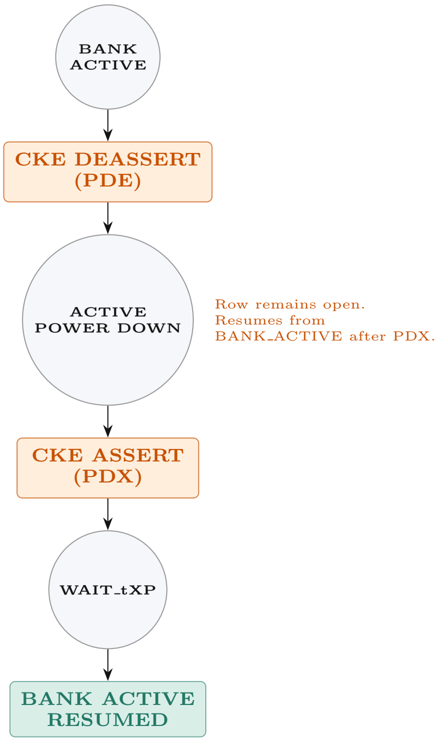

Figure 13. P2: Active power down. Open row retained. Resumes BANK_ACTIVE after tXP.

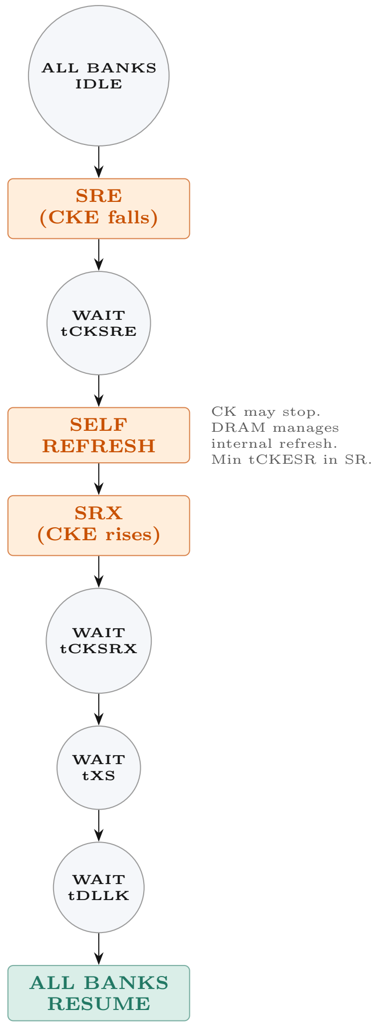

Figure 14. P3: Self Refresh entry and exit. tCKSRE, tCKSRX, tXS, tDLLK all required.

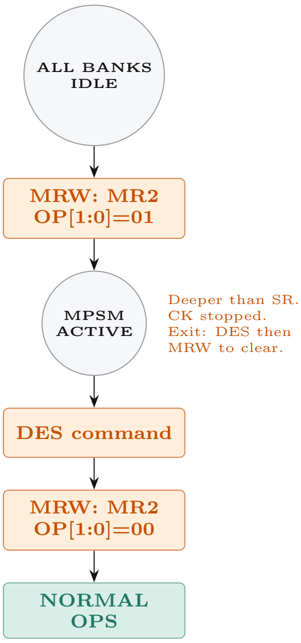

Figure 15. P4: MPSM. Deepest power state. Entry via MRW; exit via DES + MRW.

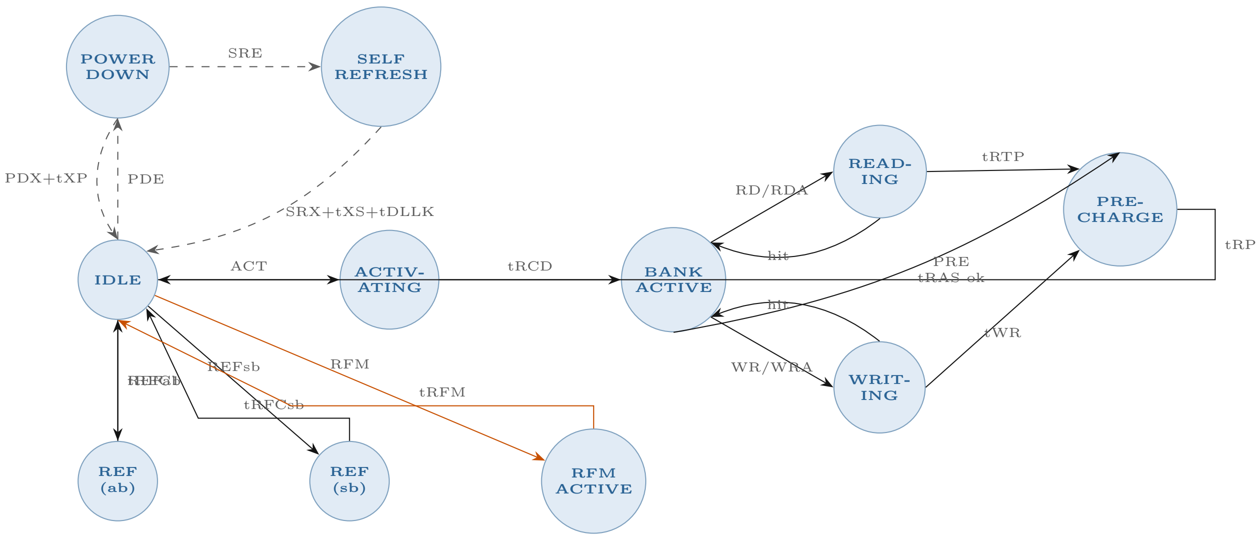

Figure 16. Full Bank FSM with all 11 states. Blue = nominal path. Orange (dashed) = power states. Amber = maintenance/RFM. Row-hit arcs keep bank in ACTIVE.